

Digital Transformation and Sustainability in Industrial Production: A Comparative Analysis of Siemens and ABB

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Group 6

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1. Introduction

In this paper, we explore how digital transformation and sustainability are shaping modern manufacturing. We start by looking at how digitalization, including technologies like Digital Twin (DT), the Internet of Things (IoT), and artificial intelligence (AI), is changing production management. We examine how these technologies are integrated into manufacturing processes and how they are enabling companies to optimize production while striving for more sustainable practices.

In addition to technological advancements, lean production strategies have become central to manufacturing practices. Traditionally, lean focuses on improving efficiency and minimizing waste. However, there is an increasing recognition of the human dimension in lean strategies. This part of our paper explores how lean practices can improve worker motivation, competency, and collaboration, contributing to a more engaged and efficient workforce.

Sustainability is another core element of digital transformation in manufacturing. Companies are trying to do more than just reduce their environmental footprint. They're also looking at social aspects, like how they treat their workers and engage with their communities. We'll talk about how companies bring sustainability into their production processes, balancing the environment and social factors.

Finally, the digital transformation of manufacturing also brings new risks and disruptions, especially in the face of external events like the COVID-19 pandemic. These disruptions have highlighted vulnerabilities in manufacturing systems and supply chains. In this paper, we discuss how companies are leveraging digitalization to address these challenges. Technologies like digital twins and real-time data analytics are helping companies anticipate disruptions, optimize their operations, and improve resilience. To ground our discussion, we analyze the strategies of **Siemens** and **ABB**, two global leaders in the industrial automation and manufacturing sectors. Siemens operates across a diverse range of industries, including energy, transportation, and healthcare, but is particularly prominent in industrial automation and digital manufacturing technologies. ABB, on the other hand, is a key player in robotics, electrification, and automation solutions, with a strong focus on serving the manufacturing, utilities, and infrastructure sectors. Both companies are at the forefront of integrating digitalization and sustainability into industrial production. By exploring their approaches, we see how digitalization not only enhances productivity but also enables adaptation to changing circumstances and drives progress toward

sustainability goals. As we go through each part of the paper — from digitalization to lean strategies, sustainability, and disruptions — we will use Siemens and ABB as real-world examples to illustrate these concepts in action.

2. State of Art

2.1 Impact of Digitalization on Production Management

The rapid advancement of information and communication technologies (ICTs) has significantly transformed the manufacturing industry. Traditional tools, such as computer-aided design (CAD), computer-aided engineering (CAE), and computer-aided manufacturing (CAM), along with methods like finite element analysis (FEA) and product data management (PDM), continue to evolve and play essential roles in industrial processes (Liu et al., 2021). At the same time, emerging technologies like big data, the Internet of Things (IoT), artificial intelligence (AI), cloud computing, edge computing, and 5G networks are revolutionizing industrial applications. These technologies enable integrating physical systems with digital technologies, which is becoming increasingly necessary to meet the growing complexity and demands of modern markets (Liu et al., 2021).

Leading companies like Siemens and ABB are actively embracing these digital technologies to enhance their manufacturing operations and meet sustainability goals. The use of digital twin technology is one key example of how these companies are leveraging real-time data to optimize production. A more detailed analysis of how Siemens and ABB implement such technologies in their production strategies will be explored further in the Discussion section.

However, despite the tremendous potential of these innovations, many industries have yet to fully leverage the strategic benefits of this integration (Liu et al., 2021). Digital tools and software used throughout the lifecycle of industrial products generate vast amounts of data. Yet, much of this data remains siloed and underutilized, reducing its effectiveness. While traditional simulations relying on static models are valuable for system verification and optimization during the planning stage, they fail to address the need for dynamic, real-time data during system runtime (Liu et al., 2021).

Recent advances in digitalization allow industries to collect and utilize real-time data, creating new opportunities for operational optimization. One such innovation is the concept of digital twins. This technology leverages real-time data to enhance system efficiency and support continuous improvement, thus enabling a more responsive and adaptive production process (Liu et al., 2021).

2.2 The Human Dimension in the Lean Production Strategy

Taiichi Ohno implemented lean thinking in manufacturing means systematic implementation of various procedures that have already been applied since the 1970s, such as TQM (Total Quality Management), just-in-time manufacturing, teamwork, and computer-based technology using a framework (Liker and Ross, 2019). It serves as a potential guide to include the human dimension of lean. Magnani, et al., in 2019, reviewed 95 articles about lean. It was found that more than half of the articles discussed the human dimensions regarding the positive and negative impacts of lean on employee conditions, such as work transformation or social effects. It has also identified an emerging interest in the human dimension as a potential lean facilitator, such as transformation policies in HR practices that facilitate lean adoption. Furthermore, the study found that lean was either described as a process-oriented strategy, focusing especially on its technical dimension, or as a people-oriented strategy, focusing primarily on its human dimension by observing how lean was implemented in organizations. The literature review also discovered that lean leads to a modification of work characteristics impacting each employee's role: employees display higher levels of competency and problem-solving capabilities, as well as increased cooperation, responsibility, and continuous motivation to improve processes.

2.3 Sustainability in Manufacturing

Based on Mary Sotos (2015), a standard for corporations requires organizations to quantify emissions from the generation of acquired and consumed electricity, steam, heat, or cooling, which are termed "scope 2" and are considered an indirect emissions source (along with scope 3) because these emissions are a result of activities of the reporting organization but occur at sources owned or controlled by another organization which are owned and managed by an electricity generator or utility. IPCC (2014), based on global emissions from 2010, also found that Scope 2 is one of the largest sources of GHG emissions globally, which is the generation of electricity and heat, which now accounts for at least a third of global GHG emissions. Both references found that electricity consumers have significant opportunities to reduce those

emissions by reducing electricity demand and increasingly play a role in shifting energy supply to alternative low-carbon resources. Unlike Scope 1 and 2, which focus on direct and indirect emissions of energy consumed by the reporting company, Scope 3 includes all other indirect emissions that occur in a company's value chain. To avoid a company double counting emissions, World Research Institute has 15 categories to classify Scope 3, which are (1) Purchase goods and services, (2) capital goods, (3) fuel-&-energy-related activities (not included in Scope 1 or Scope 2), (4) upstream transportation & distribution, (5) waste generated in operations, (6) business travel, (7) employee commuting, (8) upstream leased assets, (9) downstream transportation and distribution, (10) processing of sold products, (11) use of sold products, (12) end-of-life treatment of sold products, (13) downstream leased assets, (14) franchises, and (15) investments.

2.3 Risks and Disruptions in Manufacturing Systems: An Overview

Manufacturing systems (MSs) are vital networks that combine equipment, personnel, and processes to carry out production tasks. Over time, these systems have become more adaptable and advanced, incorporating new technologies. However, external disruptions, like the COVID-19 pandemic, have highlighted vulnerabilities in manufacturing and supply chains, leading to production delays. MSs are exposed to various risks, and while much research focuses on supply chain resilience, further exploration into the capacity of manufacturing systems to handle such disruptions is needed (Vital-Soto et al., 2024).

Risks in manufacturing and supply chains (SCs) are generally divided into two categories: operational and disruptive risks. Operational risks are more predictable and stem from uncertainties in managing demand and supply, such as machine breakdowns, transportation delays, forecasting errors, and demand fluctuations, which can be quantified through historical data. In contrast, disruptive risks result from significant external events, such as natural or man-made disasters, which can have prolonged effects and disrupt multiple tiers of the supply chain (Vital-Soto et al., 2024).

The manufacturing system life cycle consists of stages such as design, synthesis, modeling, analysis, simulation, implementation, and eventual reconfiguration or redesign.

Euler-Chelpin categorized the resource life cycle into design, planning, production, use, and disposal stages. Typically, literature divides this life cycle into four main stages (See Figure 1):

design, development/implementation, operation, and dismantling or redesign (Vital-Soto et al., 2024)

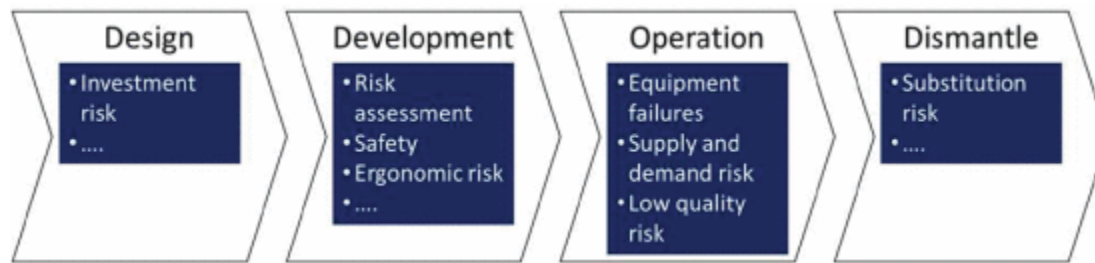


Figure 1. *Manufacturing lifecycle stages and operational risks.*

While we have established the general framework of risks and disruptions in manufacturing systems, further analysis of how these challenges are handled by the case study companies, including their strategies for mitigation, will be covered in more detail in the discussion section.

3. Discussion

3.1 Digital Twin in Practice: Industrial Applications

The Automation Pyramid is a central framework in industrial automation, consisting of five layers (see Figure 1). The lower three layers (0-2) are referred to as the control layer, including components such as the electrical engineering layer, Programmable Logic Controllers (PLCs), Distributed Control Systems (DCS), and SCADA/HMI layers. The fourth layer is the Manufacturing Execution System (MES), which guides and monitors the production process. At the top of the pyramid lies the Enterprise Resource Planning (ERP) layer, which integrates various organizational functions to improve customer support and planning (Cimino et al., 2019; ISO/IEC, 2012; Govindaraju & Putra, 2016).

The MES is responsible for overseeing the entire manufacturing process and is critical for coordinating all the steps involved in production. Some MES functions directly affect the production process, such as scheduling and quality control, while others, like resource management and traceability, provide a broader, supportive role (De Ugarte et al., 2009).

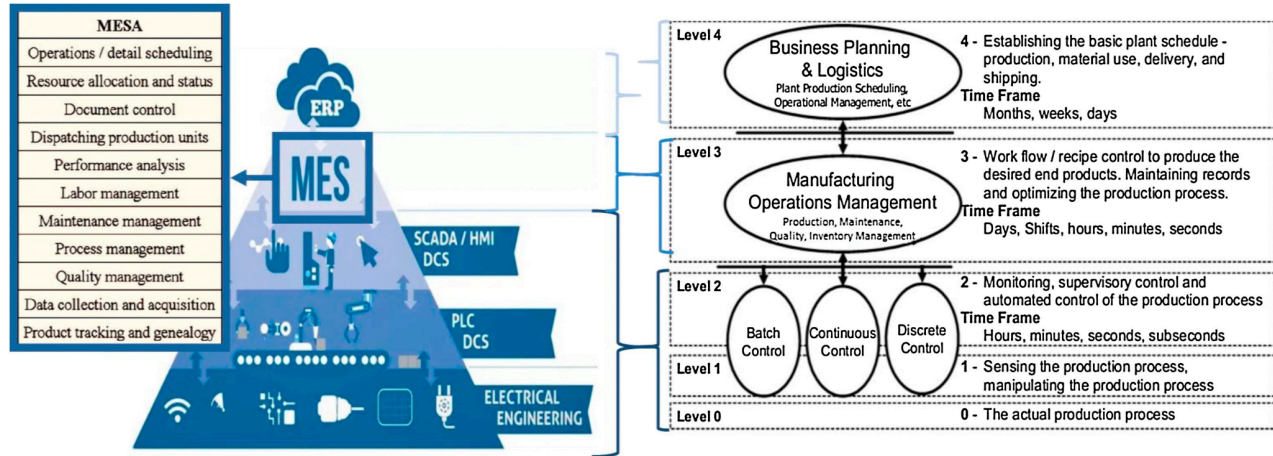


Figure 2. The automation pyramid, alongside the corresponding production management level (Govindaraju & Putra, 2016; MESA, 1997).

Kritzinger (2018) offers an insightful perspective on Digital Twin (DT) literature, categorizing it based on the interactions between physical and virtual objects. According to this view, a true Digital Twin involves bi-directional data exchange between the physical object and its virtual counterpart (from physical to virtual and vice versa). This means the virtual model can influence the control system of the physical object. Kritzinger distinguishes this full DT from two other categories: (i) The Digital Model, which lacks any interaction with the physical object, and (ii) the Digital Shadow, where only the physical object sends data to update the virtual object. In the case of the Digital Shadow, the interaction is one-way, as it involves data acquisition protocols and simulation software for the digital model. Kritzinger suggests that the Digital Shadow can evolve into a full Digital Twin once the data from the virtual model is connected to the real system, as depicted in Fig. 2 (Kritzinger et al., 2018).

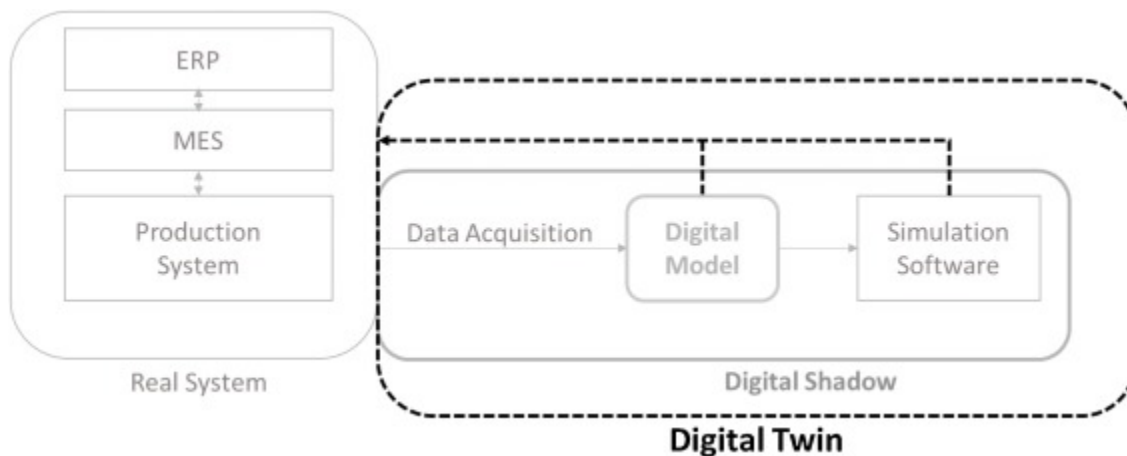


Figure 3. Digital Twin (DT) schema.

ABB's approach to Digital Twin technology in industrial applications is particularly relevant to this framework. ABB has developed a Digital Twin for its electromagnetic (EM) flowmeter products, which is based on a multiphysics finite element analysis (FEA) technique. This digital twin model mimics the real-world behavior of the flowmeter, enabling performance prediction and design optimization without the need for extensive physical testing (ABB Review, 2018). The model integrates complex physical phenomena, such as electromagnetism and fluid dynamics, to predict the flowmeter's sensitivity, accuracy, and performance under varying conditions. This predictive capability aligns with Kritzinger's perspective, as the flowmeter's Digital Twin allows for both real-time performance monitoring and predictive control, making it an example of a true Digital Twin with bi-directional data exchange.

The integration of the physical flowmeter with its virtual twin allows ABB to improve design efficiency, reduce costs, and enhance the flowmeter's performance in various operating conditions. For instance, simulations using the Digital Twin can predict how the flowmeter will behave under specific conditions, including temperature and pressure variations, which would be difficult or costly to replicate in real-world testing (ABB Review, 2018). Moreover, this digital modeling has enabled ABB to optimize component designs, such as the electromagnetic coil, to minimize material usage while maintaining or improving the product's performance. This approach highlights the potential of Digital Twin technology to enhance product development, streamline manufacturing processes, and support real-time decision-making in industrial settings. Siemens' application of Digital Twin technology plays a key role in modern industrial settings, much like ABB's approach with its Ability™ platform. Siemens introduced MindSphere in 2016, a cloud-based Industrial Internet of Things (IoT) platform that connects physical assets to the digital world, enabling predictive maintenance, real-time performance monitoring, and optimization throughout the asset lifecycle. This platform helps Siemens improve product and system performance by collecting and analyzing data from various systems, such as SLM, PLM, and MES (Annanth et al., 2021).

On the other hand, Siemens' approach to Digital Twins is broader, focusing not only on individual products but also on entire systems and production environments. This sets it apart from ABB, which tends to focus more on optimizing individual products. Siemens aims to create

Digital Twins that represent entire production lines, allowing for a more comprehensive approach to performance optimization. By integrating multiple assets into a single digital ecosystem, Siemens offers more holistic solutions that help improve operational efficiency across the entire production process (Siemens, Digital Twin technology, 2023).

Moreover, Siemens' MindSphere is built on Mendix, a low-code application development platform. This feature speeds up the creation of IoT applications, giving manufacturers the flexibility to quickly adapt to new challenges and integrate innovations. This scalability is a key advantage of Siemens' approach, allowing industries across various sectors to benefit from IoT-based solutions. In comparison, while ABB's platform also offers flexibility, Siemens' MindSphere provides more rapid deployment and integration capabilities (Siemens, MindSphere: Transforming Industries, 2023).

3.2 Lean Practice: Siemens Production System Expert Program

A study which was conducted by Emerald in 2012 identified that Siemens used a framework that could be used to improve production in all Siemens plants by implementing the principles of lean philosophy called the Siemens Production System (SPS). The finding showed that Siemens grasped that there was a shortage of experts within the firm who understood the lean philosophy well enough to implement and train their colleagues in its use. In the process, contracting with external consultants, in the long run, was proposed and rejected due to both cost and lack of long-term sustainability. The study recorded that Siemens realized a group of "home-grown" experts who could act as trainers and facilitators were needed to propagate knowledge about philosophy across the company. A tripartite training program called the Senior SPS Expert Program was developed to produce three levels of expertise: expert in the first level, senior expert in the second level, and principal expert for the third level. The firm set a nine-month target for the development and piloting of the Senior SPS Expert program, which ensures the program could be used throughout its plants. Also, training delivered in host plants combined with ongoing mentorship in participants' work environments to maintain cost-effectiveness and sustainability. Emerald (2012) found that the program result increased the number of experts from 100 to 500 at all levels combined, while productivity increased by a third-tenth at the relevant production sites with highly sustainable indicators being matched, for instance, lower energy consumption and decreased CO2 emissions.

The way that Siemens did with the program is to achieve the understanding and application of the SPS as efficiently and cost-effectively as possible in congruence with the stated lean philosophy itself. This method had the advantage of on-the-job teaching experience for “home-grown” experts during initial training and the facility to apply learning to actual work situations during the mentorship phase. Learning and teaching methods congruent with lean philosophy were developed in this program, such as training in research methods and literature searching, experiential & simulation learning, and practical exercises that enable participants to apply SPS principles in their work environments.

In comparison, ABB uses a computer-based system called ABB Ability™ Expert Optimizer (EO) to help its managers and leaders control, stabilize, and optimize industrial processes. Due to its state-of-the-art optimization technologies, the software allows managers to make the best operational decisions accurately and consistently at all times. The EO provides advanced process techniques, including linear and non-linear model predictive control, fuzzy logic, and neural networks (ABB, 2024b)

3.3 The Use of 4.0 Technologies to Overcome Environmental Sustainability

The European Union regulations strongly demand social and environmental sustainability and The Corporate Sustainability Reporting Directive (CSRD) has strengthened the regulations (European Commission, 2024). Being driven by the regulation, the code of conduct of Siemens suppliers needs to be fulfilled to ensure the effective establishment of the specified environmental, compliance, and labor standards across all countries of operations (Siemens Sustainable Procurement Report, 2023). Furthermore, Siemens (2024) claims that its direct suppliers generate only a quarter of the total upstream supply chain emissions. To achieve the net-zero target, Siemens needs to pursue the other third-fourth, beyond tier 1 and scope 2 of its supply chain, which are the 15 categories in Scope 3. At the end of 2022, Siemens SCM kicked off a platform called Supplier+s and Ctrl+s, which worked on suppliers' real-data CO2 calculations. With supplier+s, every Siemens supplier (tier 1) is expected to nominate their suppliers (tier 2) to complete the assessment in an easy, user-centric, and secure way. With the platforms, Siemens can account for and control category numbers 4, 5, 8, and 10 of Scope 3 mentioned in section 2.3. For instance, in transportation and distribution of products and services purchased, including inbound and outbound logistics, Supplier+s is where the carbon accounting module comes in, based on readily available data spending and industry statistics, it can easily

quantify the carbon footprint and immediate insight into the structure of Scope1, 2, and 3 emissions to understand emission hotspots and therefore strategically prioritize decarbonization efforts for maximum impact (ctrl+s, 2024). However, getting informative and transparent data in the supply chain from suppliers is not that simple and straightforward. Many companies do not want to share their data even with their customers due to data security, privacy risks, and loss of competitive advantages (Kshetri, 2024). BlueVoyant, a cybersecurity company, (2023) reveals more than one-fifth (26%) of organizations increase in reported negative impacts from supply chain cyber breaches, disrupting operations, and highlights the growing threat. To cope with these challenges, Siemens had some historical cases that can be used below:

3.3.1 Siemens Cybersecurity

As a business-to-business (B2B) company, there is a cyber security technology Siemens has implemented with Amazon called Siemens Cybersecurity by using AWS Machine Learning where it can handle more than 60000 threats per second. At this point, They had launched Siemens Cyber Defense Center, to secure its customers from malware, viruses, and various other forms of security threats of cybercrime (Pospisil, 2019). Later, they also gave assurance that their AI-driven cybersecurity platform was easily among the top and most needed in the world with the present increasing rate of cybercrimes (Annanth, et al., 2021). Based on this case, Siemens can use this platform for their customers and implement AI-driven cybersecurity to overcome data security and privacy risk issues for their suppliers on the Supplier+s and Ctrl+S platforms. If that is so, Siemens can achieve their net-zero target in scope 2 and 3 of their supply chain.

3.3.2 Totally Integrated Automation (TIA)

Industry 4.0 plays a significant role in the development, which was brought to life through Siemens' strengths in industrial automation. In the 1990s, Siemens launched the "Totally Integrated Automation" (TIA) Portal, which enabled companies to coordinate their production operations and interconnect hardware with software. TIA focused on increased speed, flexibility, and productivity by developing innovative simulation tools, transparent plant operation, and integrated engineering in a whole package. Then, cloud solutions as a networked workflow maintain the consistency of total data, which contributes to the overall digital workflow. For instance, in a typical case study, Siemens TIA had been used by Audi AG, a German automobile manufacturer, when they initiated the production of the Audi A8. The result is a drop of more than one-tenth (10%) in the engineering costs. Also, utilizing TIA had the major advantage in

handling efficiency and reducing downtime by assuring information being delivered to central control systems by individual control systems (Annanth, et al., 2021). The value that can be learned from this case is how to implement such a TIA system not only for customers but also for Siemens' suppliers to manage the loss of competitive advantages on the supplier side. It is beneficial for Siemens not only to grow together with the suppliers but also to get reliable & transparent information and the real production operations & conditions from the suppliers of scopes 2 and 3. On the other hand, the suppliers have an advantage in sharing information with Siemens because they will have Siemens support by the TIA system.

On the other, The ABB Supplier Code of Conduct (SCoC) outlines its expectations on principles contained within the International Bill of Human Rights, the ILO Core Conventions on Labor Standards, and all applicable laws and principles in the areas of human rights and decent work. ABB has also implemented the Sustainable Supply Base Management (SSBM) approach to improve the sustainability performance of the suppliers, especially those exposed to sustainability risks addressing sustainability topics and performance at each stage of supplier life cycle management, as part of the audit approach, which covers issues in six main categories: general management, human and labor rights, social benefits, health, safety, and the environment (ABB, 2024a).

3.4 Managing Disruptions in Manufacturing: Strategies and Risk Mitigation

During the operation phase of manufacturing systems (MSs), several risks, including material shortages, resource failures, and operator unavailability, may arise. To manage these disruptions, various methods have been suggested. Darmoul et al. (2013) proposed an immune-based framework using multi-agent systems to monitor and handle unexpected events. Bayar et al. (2016) extended this idea, creating a knowledge-based system for detecting and reacting to disruptions. Additionally, scheduling and rescheduling strategies, such as those outlined by Vieira et al. (2003) and Tanimizu et al. (2006), focus on reacting to operational disturbances through adaptive scheduling methods. (Vital-Soto et al., 2024)

To tackle operational risks, manufacturing systems employ strategies like flexibility and redundancy. Flexibility involves approaches such as scheduling, routing, and adjusting capacity to handle uncertainties. Redundancy ensures backup resources are available when primary resources fail, though it is costly and typically reserved for critical systems. While many studies focus on reactive strategies like planning and control, which address risks once they occur, there

is growing interest in proactive and predictive methods, driven by the use of data from sensors. These strategies have the potential to significantly enhance system performance but are still evolving (Vital-Soto et al., 2024).

During the COVID-19 pandemic, ABB faced significant disruptions due to a variety of challenges, including supply chain delays, labor shortages, and restrictions on in-person operations. To address these challenges, ABB leveraged its digital capabilities, which were central to maintaining operational continuity. A key part of their strategy was the use of AI and predictive maintenance technologies. These tools allowed ABB to remotely monitor production lines, predict failures, and minimize downtime. By integrating their production systems with the Ability platform, ABB was able to collect real-time data and provide predictive insights, enabling swift responses to emerging issues. The company also embraced digital transformation and automation, which helped them stay resilient during the pandemic. Furthermore, ABB adopted remote services and virtual collaboration tools to ensure operations could continue even with restrictions on physical presence. These actions were part of ABB's broader strategy to increase flexibility and resilience in their manufacturing processes (ABB Annual Report, 2020).

Similarly, Siemens faced similar disruptions during the pandemic, particularly with supply chain challenges and labor shortages. In response, Siemens focused on a digital-first strategy to mitigate these risks. The company relied heavily on its established smart factory infrastructure and digital twins to monitor and optimize production. Siemens used its MindSphere platform, which is an IoT (Internet of Things) solution, to track equipment and production systems remotely. Through real-time data analytics, Siemens could predict maintenance needs and prevent potential downtime, optimizing production workflows. Like ABB, Siemens' digital solutions provided a significant advantage, enabling them to address disruptions more effectively and maintain operational stability (Siemens Annual Report, 2021).

Both ABB and Siemens used digital tools to manage disruptions during the pandemic, focusing on predictive maintenance and remote monitoring. ABB relied on its Ability platform for real-time insights, while Siemens utilized MindSphere and digital twins to optimize operations. Both companies leveraged data analytics to improve efficiency and minimize downtime.

However, we argue that such advanced digital solutions, while effective for large corporations like Siemens and ABB, may not be realistic for small and medium-sized enterprises (SMEs) due to the significant financial investments required. According to the National Institute of Standards

and Technology (NIST), implementing technologies like digital twins involves substantial upfront costs for infrastructure, training, and integration, which can be prohibitive for SMEs (NIST, 2021). Additionally, the Organisation for Economic Co-operation and Development (OECD) highlights that SMEs often face constraints in digital infrastructure and financial resources, which limits their ability to adopt these transformative technologies (OECD, 2023).

While these technologies are viable solutions for addressing disruptions and enhancing resilience, the cost barrier makes them less accessible to SMEs. This disparity emphasizes the need for scalable, cost-effective solutions that cater to smaller enterprises. Policymakers, industry leaders, and technology providers must collaborate to bridge this gap through initiatives like subsidies, shared digital infrastructures, or simplified technology packages to ensure that SMEs are not left behind in the digital transformation journey.

4. Conclusion

In conclusion, the adoption of digital technologies like Digital Twins, IoT, and Lean production is transforming industries in big ways. These technologies are making production processes more efficient, reducing waste, and helping companies become more sustainable. By looking at companies like Siemens and ABB, we can see how these innovations are being used to improve operations and cut down on environmental impact.

For example, Digital Twins are helping companies optimize production by improving maintenance, saving energy, and making the design process more efficient. Siemens, through its MindSphere platform, is able to monitor and adjust its operations in real time, while ABB uses Digital Twins to monitor products and lower costs. These examples show how digitalization is helping companies become more sustainable in their operations.

Additionally, Lean production methods are becoming more common in many industries to create more flexible and responsive production lines. As we discussed earlier, the human dimension factor is just as important to consider. People play a crucial role in making Lean strategies work effectively, as they ensure the adaptability and continuous improvement that Lean production requires. Further, despite advancements in this industry, we discussed how the COVID-19 pandemic could cause disruptions even for giant companies like ABB and Siemens. However, the integration of Digital Twin technology and other tech solutions has helped these companies thrive through such challenges, enabling them to maintain operations and adapt to rapidly changing conditions.

To conclude, it can be seen that advanced digitalization is not only a key driver for improving operational efficiency and sustainability but also a crucial enabler for companies to navigate future disruptions and thrive in an increasingly volatile and fast-changing global environment.

References

- [1] Magnani, F., Carbone, V. and Moatti, V. (2019). The human dimension of lean: a literature review. *Supply Chain Forum: An International Journal*, 20(2), pp.132–144. doi <https://doi.org/10.1080/16258312.2019.1570653>.
- [2] Sotos, M. (2015). *GHG Protocol Scope 2 Guidance*. GHG Protocol Team, World Research Institute. [online] Available at: <https://ghgprotocol.org/sites/default/files/2023-03/Scope%202%20Guidance.pdf>.
- [3] Barrow, M., et al., (2013). *Technical Guidance for Calculating Scope 3 Emissions Supplement to the Corporate Value Chain (Scope 3) Accounting & Reporting Standard*. [online] Available at: https://ghgprotocol.org/sites/default/files/2023-03/Scope3_Calculation_Guidance_0%5B1%5D.pdf.
- [4] Emerald (2012). Linking theory and practice in Siemens' new training initiative. *Strategic Direction*, Emerald Journal, 28(11), pp.12–13. doi:<https://doi.org/10.1108/02580541211278554>.
- [5] European Commission (2024). *Corporate sustainability reporting*. [online] European Commission. Available at: https://finance.ec.europa.eu/capital-markets-union-and-financial-markets/company-reporting-and-auditing/company-reporting/corporate-sustainability-reporting_en.
- [6] Siemens (2023). *Sustainable Procurement Supply Chain Management Version 1.2, Oktober 2023*. [online] Available at: <https://assets.new.siemens.com/siemens/assets/api/uuid:2ea3d787-8ee1-4fa5-b0a9-051b8c16974b/Siemens-Sustainable-Procurement.pdf>.
- [7] Siemens (2024). *Carbon Reduction @ Suppliers*. [online] siemens.com Global Website. Available at: <https://www.siemens.com/global/en/company/sustainability/sustainablesupplychain/carbon-reduction-suppliers.html> [Accessed 3 Dec. 2024].
- [8] Ctrl+s (2023). *The new Carbon Accounting Module in supplier+s is live*. [online] Ctrl-s.io. Available at: <https://ctrl-s.io/blog/the-new-carbon-accounting-module-in-supplier-s-is-live/> [Accessed 3 Dec. 2024].
- [9] Kshetri, N. (2024). *Supply Chain Trust*. [online] Lib.kth.se. Available at: <https://ieeexplore-ieee-org.focus.lib.kth.se/stamp/stamp.jsp?tp=&arnumber=8675705> [Accessed 3 Dec. 2024].
- [10] Blue Voyant (2023). *BlueVoyant Research Reveals Impact from Supply Chain Cyber Attacks*. [online] BlueVoyant. Available at: <https://www.bluevoyant.com/press-releases/organizations-continue-to-brace-for-impact-from-supply-chain-cyber-attacks> [Accessed 3 Dec. 2024].
- [11] Pospisil, J. (2019). Siemens handles 60,000 cyber threats per second using AWS machine learning [online]. Available at: <https://aws.amazon.com/solutions/case-studies/siemens-cybersecurity>
- [12] ABB (2024a). *ABB Sustainable Supply Base Management Program*. [online] ABB Group. Available at: <https://global.abb/group/en/about/supplying/responsible-sourcing>.
- [13] ABB (2024b). *Ability™ Expert Optimizer - Cement Mill Optimization Solutions | ABB*. [online] Cement. Available at: <https://new.abb.com/cement/systems-and-solutions/advanced-process-control/abb-ability-expert-optimizer-cement> [Accessed 3 Dec. 2024].
- [14] Siemens Digital Industries Software (2024). *Balance costs and the carbon footprint | Siemens Software*. [online] Siemens Digital Industries Software. Available at: <https://resources.sw.siemens.com/en-US/case-study-siemensag/> [Accessed 3 Dec. 2024].

- [15] Business Wire (2021). New ABB study on industrial transformation unveils critical relationship between digitalization and sustainability.
- [16] Siemens & MIT Technology Review (2023). The Emergent Industrial Metaverse.
- [17] Vuković, M., Jorg, O., Hosseinifard, M., & Fantoni, G. (2022). Low-Cost Digitalization Solution through Scalable IIoT Prototypes. *MDPI Electronics*, 12(17), 8571. [online] Available at: <https://www.mdpi.com/2076-3417/12/17/8571>
- [18] Product Lifecycle Management Pre-Study: An Investigative Case at Industrial Automation, ABB AB. 2020. [online] Available at: <https://odr.chalmers.se/items/0108d1e7-cbca-4dc3-8974-c362abc29d3a>
- [19] Industry IoT Case: Connected Services at ABB Robotics. I-Scoop. Available at: <https://www.i-scoop.eu/internet-of-things-iiot/industrial-internet-things-iiot-saving-costs-innovation/iiot-abb-robotics-case/>
- [20] Jeske, T., Würfels, M., & Lennings, F. (2021). Development of Digitalization in Production Industry – Impact on Productivity, Management and Human Work. *ScienceDirect*. [online] Available at: <https://www.sciencedirect.com/science/article/pii/S1877050921004348>
- [21] Liu, M., Fang, S., Dong, H., & Xu, C. (2021). Review of digital twin about concepts, technologies, and industrial applications. *ScienceDirect*. Available at: <https://www.sciencedirect.com/science/article/pii/S0278612520301072#bib0005>
- [22] Cimino, C., Negri, E., & Fumagalli, L. (2019). Review of digital twin applications in manufacturing. *ScienceDirect*. Available at: <https://www.sciencedirect.com/science/article/pii/S0166361519304385>.
- [23] ABB Review (2018). ABB's Electromagnetic flowmeter digital twin drives performance. Available at: https://www.researchgate.net/profile/Vinay-Kariwala/publication/326177993_ABB%27s_Electromagnetic_flowmeter_digital_twin_drives_performance/links/5c29e4ca458515a4c702ab17/ABBs-Electromagnetic-flowmeter-digital-twin-drives-performance.pdf.
- [24] Uhlemann, T. H.-J., Lehmann, C., & Steinhilper, R. (2017). The Digital Twin: Realizing the Cyber-Physical Production System for Industry 4.0. *Procedia CIRP*, 61, 1–6. Available at: <https://www.sciencedirect.com/science/article/pii/S2212827116313129>.
- [25] Melesse, T. Y., Di Pasquale, V., & Riemma, S. (2020). Digital Twin Models in Industrial Operations: A Systematic Literature Review. *Procedia Manufacturing*, 51, 994–1001. Available at: <https://www.sciencedirect.com/science/article/pii/S2351978920306491>.
- [26] Vital-Soto, A., & Olivares-Aguila, J. (2024). Manufacturing Systems for Unexpected Events: An Exploratory Review for Operational and Disruption Risks. *IEEE Access*, 12, 10237110. Available at: <https://ieeexplore.ieee.org/abstract/document/10237110>
- [27] Siemens. (2020). MindSphere@SCM: Full speed ahead to digitalization. *Siemens Blog*. Available at: <https://blog.siemens.com/2020/10/mindspherescm-full-speed-ahead-to-digitalization/>
- [28] Siemens. (n.d.). Siemens Digital Twins with potential. *Siemens*. Available at: <https://www.siemens.com/global/en/company/stories/industry/the-digital-twin.html>
- [29] Demartini, M., Evans, S., & Tonelli, F. (2019). Digitalization Technologies for Industrial Sustainability. *Procedia Manufacturing*, 33, 700-707. Available at: <https://www.sciencedirect.com/science/article/pii/S2351978919305098>
- [30] Liu, M., Fang, S., Dong, H., & Xu, C. (2021). Review of digital twin about concepts, technologies, and industrial applications. *Computers and Industrial Engineering*, 148, 106784. Available at: <https://www.sciencedirect.com/science/article/pii/S0278612520301072#bib0005>
- [31] Kishorre Annanth, V., Abinash, M. and Rao, L.B. (2021). Intelligent manufacturing in the context of industry 4.0: A case study of siemens industry. *Journal of Physics: Conference Series*, 1969(1), p.012019. doi:<https://doi.org/10.1088/1742-6596/1969/1/012019>.
- [32] Liker, J. and Ross, K., (2017). The Toyota way to service excellence. *Lean transformation in service organizations*.
- [33] Liker, J. and Ross, K., (2017). The Toyota way to service excellence. *Lean transformation in service organizations*.

[34] National Institute of Standards and Technology (NIST), *Economics of Digital Twins: Costs, Benefits, and Economic Decision Making*. Available at: [Economics of Digital Twins: Costs, Benefits, and Economic Decision Making | NIST](#)

[35] Organisation for Economic Co-operation and Development (OECD), *SMEs Going Digital: Policy challenges and recommendations*. Available at: [SMEs Going Digital: Policy challenges and recommendations](#)